

SWEN 563 / CMPE 663 / EEEE 663 — Real-Time & Embedded Systems

Project 1 — PC-1000 "PaceCheck": Pacing-Interval Analyzer for Pacemaker Production QA

Pocket Instrumentation Suite · Instrument #1 (bare metal)

Assigned	Wed Sep 2, 2026 (Session 4)
Demo (in studio)	Wed Sep 16, 2026 (Session 7). Overflow slots Mon Sep 21 (Session 8)
Memo + code due	Wed Sep 16, 2026 (Session 7), at the in-class demo
Weight	10% of course grade
Teams	2 students
Platform	STM32WB5MM-DK, bare-metal register-level C (no HAL)
Bench equipment	Analog function generator (any make) + digital storage oscilloscope
Supporting lectures	L4 GPIO · L5 Clocks/SysTick · L6 Interrupts/NVIC · L7 Timers & input capture (Sep 2) · L8 UART (Sep 14)

1. Scenario

Your customer is **Kestrel Rhythm Management**. The company makes implantable cardiac pacemakers. A pacemaker must deliver a pacing pulse at exactly the programmed moment. Kestrel applies one acceptance rule to each device that leaves the production line. The rule is **1000 consecutive pace-to-pace intervals, each in $\pm 100 \mu\text{s}$ of the programmed interval**, with the measurement record archived as evidence.

The acceptance rule has a cause. The initial end-of-line test at Kestrel calculated the average pacing rate for one minute. That test accepted devices that sometimes gave a pulse **8 ms late**. The average value was correct, but the individual intervals were not. The lot almost caused a recall. An average value hides one late pulse in 1000, and a per-interval histogram shows it.

You are the firmware engineer for the instrument that enforces the rule. The instrument is the **PaceCheck PC-1000**, a bench analyzer for the Kestrel production line. At your laboratory bench the **function generator acts as the pacemaker**, and the **oscilloscope gives the ground truth**. The provided LOOPBACK stimulus (§6) lets you build and test the capture engine at home with no equipment.

What the PC-1000 actually does — the core function

This project has one core function.

Timestamp 1001 consecutive rising edges with a $1 \mu\text{s}$ hardware timer. Calculate the 1000 intervals between the timestamps. Put each interval into a $1 \mu\text{s}$ -wide histogram bin. Give the PASS verdict only when each one of the intervals is in the acceptance window $[E-T, E+T]$.



The measurement pipeline has five steps.

- 1 **Capture.** Each rising edge on the SENSE pin makes the timer hardware latch the free-running 1 MHz count into a register (TIM2 input capture). The hardware makes the timestamp, and your code collects the latched values. A software loop that polls a GPIO does not have sufficient speed, and it is not stable at 1 μ s resolution. This limit is the cause of input capture.
- 1 **Difference.** Consecutive timestamps subtract to give each interval. Calculate the first interval from two edges that the timer captured in the current acquisition (R2). A stale power-on timestamp causes one garbage interval, and demo step D3 finds it.
- 1 **Bin.** Each interval increments one cell of an array that spans $[E-T, E+T]$ with one cell for each microsecond. The instrument counts an interval below the window as OUT_LOW and an interval above the window as OUT_HIGH. An acquisition of N intervals must account for exactly N in the report (R5).
- 1 **Judge.** PASS means that all N intervals are in the window. One outlier gives FAIL (R8).
- 1 **Report.** The report gives the non-empty bins, the statistics, and the verdict (R5–R8). This example shows the necessary content, and you select the layout.

```

CAL-VERIFY E=1000 T=50 N=1000
 998    41
 999   233
1000   454
1001   236
1002    36
TOTAL: 1000 in-window + 0 OUT_LOW + 0 OUT_HIGH = 1000
min 998  max 1002  mean 1000.0  sd 0.9  outtol 0  rate 60000.4 ppm
worst early -2.0 us  worst late +2.0 us  margin to limit 48.0 us
VERDICT: PASS

```

A few adjacent bins, and not one bin, are the correct result on this bench. The cycle-to-cycle jitter of the analog generator spans a few microseconds, and your 1 μ s instrument resolves it. A histogram with one bin is suspicious here. Know the cause before your demo, because the demo Q&A asks this question.

The histogram gives more information than a mean value or a min/max pair. The shape of the histogram gives the diagnosis. One narrow bin shows a crystal-locked pace scheduler. A Gaussian smear of bins shows oscillator noise or supply noise. Two separated clusters show a firmware mode-switch defect. One bin far from the other bins shows the late-pulse defect that almost caused the recall.

A mean value cannot show the difference between these four shapes. A min/max pair cannot show the difference between noise and a mode-switch defect. The demo (§9) makes you cause the first three shapes live at the bench. The three bench stimuli are the jitter of the analog generator, a slow turn of the frequency dial, and one fast dial step. The late-pulse shape comes from the field, and the R5 OUT_HIGH tally is what preserves it when it occurs.

This project builds the measurement engine and qualifies it at bench rates. True pacing is slow. Pacemakers pace at 30–180 ppm (paces-per-minute), so 1000 intervals at 60 ppm take a 17-minute soak. The factory acceptance procedure at true pacing rates reuses the engine that you qualify here, and that procedure is a documented overnight-rack task outside this project. Your scope is **CAL-VERIFY**: prove the 1 μ s measurement engine on bench square waves from **500 Hz to 10 kHz**, with the oscilloscope as the reference. An acquisition of 1000 intervals takes 0.1 s at 10 kHz and 2.0 s at 500 Hz, and the demo uses this speed.

2. Learning objectives

At the end of this project you can do these tasks.

- 1 Configure **RCC, GPIO, and TIM2 input capture** fully by direct register access with read-modify-write discipline. Use the reference-manual documentation (RM0434) and do not use HAL calls.

- 1 Build a **1 μ s hardware timebase** with prescaler math and a 32-bit free-running counter. Use **timer input capture** to timestamp external edges without software-induced error.
- 1 Make firmware that reports honestly. The firmware accounts for every measured interval in its report, tolerates out-of-tolerance input, and offers the next acquisition without a reset (R5, R11).
- 1 Isolate **time-critical capture** from **non-critical I/O** such as the console and the display. This project gives the first example of the real-time discipline that Part 2 formalizes.
- 1 Use the bench instruments as an engineer uses them. Configure the signal generator as a specified stimulus. Use the oscilloscope as the **ground truth** that measures the stimulus independently of your instrument.

3. The instrument — modes of operation

Mode	Purpose
POST	Power-on self-test. The instrument makes sure that DUT edges are on the SENSE input in 100 ms before it shows the operator console. The check answers two questions. Does the fixture pace, and is the clip lead on? If the check finds no edge, the console gives a retry loop. The 100 ms window is 50 periods at the 500 Hz low end of the band.
CAL-VERIFY	Instrument qualification. The instrument measures 1000 consecutive intervals of a bench square wave in the 500 Hz to 10 kHz band, and the demo centers on ~ 1 kHz. It uses TIM2 32-bit input capture at 1 μ s resolution, and the oscilloscope is the reference. The report gives a histogram with one bin for each μ s. The report also gives the OUT_LOW and OUT_HIGH tallies, the summary statistics, and the PASS/FAIL verdict. The defaults are $E = 1000 \mu$ s and $T = \pm 50 \mu$ s.
CONFIG	UART operator console (provided driver). The operator can view and set E and T. The operator can start an acquisition, and can do a new acquisition with the same or new settings without a reset. All output is clean in the PuTTY-log, because the PuTTY-log is the archived inspection record.
Graduate requirement	DUTY-CYCLE histogram with the interval histogram. This extension is required for a 663-section student (§8).

4. Functional requirements

The requirement language follows RFC 2119. **SHALL** is mandatory, and the team must demonstrate each SHALL at the bench. R3 is different, because R3 is a source-code property: the code submission verifies it.

R1 — POST. At power-up the PC-1000 SHALL do a power-on self-test. The self-test SHALL make sure that a minimum of one rising edge occurs on the SENSE input (Arduino **D5 = PB10**) in **100 ms**. On a timeout the PC-1000 SHALL report NO DUT SIGNAL (no edge within 100 ms) on the UART console and on the OLED. The PC-1000 SHALL then offer a retry through a console keypress, again and again, without a power-cycle. On success the PC-1000 SHALL show the operator console.

The 100 ms window is 50 periods at the 500 Hz low end of the band, so a present signal always passes POST.

R2 — Capture engine. The PC-1000 SHALL measure consecutive rising-edge-to-rising-edge intervals with **TIM2 32-bit input capture, channel 3, on PB10 (AF1)**. The PC-1000 SHALL measure at **1 μ s resolution** with a 64 MHz timer kernel clock and **PSC = 63**. The PC-1000 SHALL miss no edge and SHALL count no edge two times,

for pulse rates from **500 Hz to 10 kHz**. The PC-1000 SHALL calculate the first interval from two edges that the timer captured after the start command. A timestamp latched before the acquisition start SHALL NOT contribute to an interval. A timestamp from POST is one example. The 32-bit counter at 1 MHz spans ~71 minutes, and you must know this number.

R3 — Bare metal & pin ownership. Student code SHALL configure all RCC, GPIO, TIM2, and NVIC registers for POST and the measurement engine. The student code SHALL use direct register access with read-modify-write discipline. The only permitted black-box code is the provided console, OLED, RGB-LED, and clock drivers (§6). These drivers **exclusively own PA1, PA7, PH0, PH1, PC8, PC9, PB6, PB7**. Student code SHALL NOT reconfigure those pins (refer to the risk statement in §7.3).

R4 — Acquisition configuration. The operator console SHALL display the current expected interval **E** and the tolerance **T** (acceptance window **[E–T, E+T]**). The console SHALL accept new values in these limits.

- $100 \leq E \leq 2,000 \mu\text{s}$ (the 500 Hz to 10 kHz band)
- $10 \leq T \leq 500 \mu\text{s}$
- $T < E$

The interval count is fixed at **N = 1000**, because a qualification always uses the full count. The acquisition takes 0.1 s at 10 kHz and 2.0 s at 500 Hz.

The T limit gives room for the jitter and the drift of the analog bench source at the 500 Hz end of the band. A true pacemaker production line uses a much tighter limit on much better hardware.

The CAL-VERIFY defaults SHALL be **E = 1000, T = 50**.

The console SHALL reject an out-of-range entry or a malformed entry with an explanatory message and a re-prompt. The console SHALL echo the active window before each acquisition.

R5 — Histogram report. The instrument SHALL put an in-window interval into a **1 μs bin in the range [E–T, E+T]**. The instrument SHALL count an out-of-window interval as **OUT_LOW** (a pace before the window) or **OUT_HIGH** (a late pace). After each acquisition the console SHALL give **only the non-empty bins** in ascending interval order, as two columns `interval_μs` count. The console SHALL then give the OUT_LOW tally and the OUT_HIGH tally. The console SHALL end the report with a totals line, for example TOTAL: 990 in-window + 6 OUT_LOW + 4 OUT_HIGH = 1000. The sum SHALL be **exactly N**. The TA compares the totals line with the printed bins in your PuTTY-log after the demo.

The instrument must calculate the totals line correctly, and the TA can find a hardcoded value.

R6 — Summary statistics. The report SHALL include the minimum interval, the maximum interval, and the mean interval. The report SHALL give the mean to **0.1 μs** , and the firmware SHALL calculate it with integer arithmetic. The report SHALL include the **sample standard deviation to 0.1 μs** and the out-of-tolerance count, which is the OUT_LOW + OUT_HIGH sum from R5. The report SHALL include the equivalent pacing rate $\text{rate_ppm} = 60,000,000 / \text{mean_}\mu\text{s}$ to 0.1 ppm.

Each rate in the 500 Hz to 10 kHz band is a calibration rate and not a clinical rate. Give this statement on the console banner.

R7 — Worst-case pacing error. The summary SHALL give the worst-case deviation from the programmed interval in acceptance vocabulary. The summary SHALL give the worst-early value ($\text{min} - E$) and the worst-late value ($\text{max} - E$) in signed μs . The summary SHALL also give the remaining **margin to the $\pm T$ limit**. A Kestrel line operator accepts or rejects a device from the margin, and not from the raw minimum and maximum.

R8 — Verdict indication. The RGB LED (provided driver) SHALL show **blue** during an acquisition. The RGB LED SHALL show **green (PASS)** if and only if all N intervals are in-window. The RGB LED SHALL show **red (FAIL)** if $\text{OUT_LOW} + \text{OUT_HIGH} > 0$. The verdict SHALL latch until the next acquisition. The OLED and the console SHALL also show the verdict with the tallies.

R9 — Display policy. During an acquisition the OLED SHALL display a static ACQUIRING screen with **no per-sample redraws**, so display I/O cannot perturb the capture at kHz rates. After each acquisition the OLED SHALL show the verdict, the minimum, and the maximum.

Record this display decision in your report, because it is the main lesson of R9 and R10.

R10 — Deferred I/O. The instrument SHALL NOT transmit on the UART between the first captured edge and the last captured edge of an acquisition. The main loop SHALL issue all OLED updates, and the capture path SHALL NOT issue an OLED update. The demo makes sure of compliance from the behavior of the instrument. Demo D8 does the R5 sum-equals-N check at a **10 kHz** high rate, where an edge occurs each 100 μ s.

The demo asks why the instrument must not transmit during a capture, so prepare an answer.

R11 — Re-run. After each report the console SHALL offer a new acquisition with the same settings or with changed settings. The console SHALL offer these selections indefinitely, without a new start of the firmware.

R12 — Throughput. While the DUT signal continues, a CAL-VERIFY qualification cycle SHALL complete in **5 s** of the start command. The cycle includes the capture, the histogram, the statistics, and the verdict. The daily line-clearance procedure then costs the operator less than ten seconds.

Scope note. POST (R1) is the only signal-presence check in this project. *A signal loss in the middle of an acquisition is not part of this project. The course does not specify that behavior, and the course does not grade it.*

5. Design constraints

- 1 **Register-level C.** Do not use the STM32 HAL, CubeMX-generated code, or CMSIS convenience functions other than the register definition headers. You can use the `stm32wb55xx.h` register structs, and the course recommends them.
- 1 **Read-modify-write.** Configure only the bits that you own, with `|=`, `&=~`, or masked writes. A wholesale write to a shared register such as `GPIOA->MODER = ...` breaks the pins of the provided drivers and stops your demo before the end (§7.3).
- 1 **The course permits polling for the capture.** Poll `TIM2->SR CC3IF` as the provided *Timer Setup for Capture* handout gives it, or use a capture interrupt. The graduate extension mandates a capture interrupt for both-edge capture (§8).

Know the interaction at the 10 kHz point (D8). An OLED text write takes hundreds of μ s, and an edge occurs each 100 μ s. A polled capture loop cannot coexist with mid-acquisition display writes at that rate. R9 mandates the static ACQUIRING screen because of this limit.

- 1 **Integer arithmetic** for all statistics (R6/R7). Do not use `float` or `double` in the measurement path or in the reporting path. Fixed-point arithmetic on integers is part of the test. Be careful of the second traditional trap, which comes after the PSC+1 trap. A 32-bit sum of squared intervals overflows inside the legal configuration range. At the 500 Hz end, 1000 intervals near the 2500 μ s window top give $\sum x^2 \approx 6.3 \times 10^9$, which is $\sim 1.5 \times \text{UINT32_MAX}$.

Use 64-bit accumulators (`uint64_t`) or mean-centered sums. A 32-bit accumulator gives a garbage σ at the 500 Hz demo steps (D6/D7), but it passes D3 at 1 kHz silently. That asymmetry is deliberate.

- 1 **Code structure counts** (refer to the grading table). Isolate the hardware access behind small functions. Do not use copy-paste-and-edit-slightly blocks. Use meaningful names. Give a header comment at the top of each source file.

- 1 **The prescaler trap, in writing.** The counter clock is $f_{CK} / (PSC + 1)$. At 64 MHz, **PSC = 63** gives a 1 MHz tick, and PSC = 64 does not. Many students before you shipped the off-by-one error, and demo step D4 finds it. Derive the value and do not inherit it.

6. Provided starter code (black boxes)

The course gives you working drivers, so your two weeks go into the registers, the capture, and the histogram engine. The drivers keep you away from string parsing and display plumbing. Do not change the drivers. Do not send the drivers as your work. The driver pins are off-limits (R3).

The one **shell** marked *you complete it* obeys a different rule. You write the fenced STUDENT sections in that shell, and you send them. The plumbing around those sections is not your work, and the grade does not include it.

Module	What it gives you
clock.c/h	64 MHz SYSCCLK bring-up.
console.c/h	USART1 virtual COM port (115200 8N1) with a <code>printf</code> retarget and a line-read and tokenize helper . The E/T entry validation (R4) should be logic and not <code>strtok</code> work.
oled.c/h	Text API for the 0.96" 128×64 OLED (SSD1315 on SPI1). The per-line text writes are cheap. Use this API from your main loop only (R9/R10).
rgb_led.c/h	<code>rgb_set(r, g, b)</code> for the RGB LED. A TI TLC59731 drives the LED through a single-wire pulse-coded protocol on PA7. That protocol is the cause of the provided driver and of the protection of PA7.
Timer Setup for Capture (PDF)	Register-by-register walkthrough of TIM2 input capture: RCC APB1ENR1 enable → PSC load + EGR update force → CCMR2 channel 3 input select/filter → CCER enable → CR1 CEN → poll SR CC3IF → read CCR3 (which clears the flag). The handout ships with the assignment as the register-by-register companion to L7, which the course teaches on the day that it assigns P1 (Sep 2).
loopback.c/h	The LOOPBACK self-stimulus, complete: a SysTick-paced square wave on PD12 (Arduino D2) that you jumper to D5. It validates your capture engine at home, with no bench equipment. Its duty cycle is ~50%, so it also gives the 663 duty-cycle extension a known at-home reference (§8).
menu.c/h (<i>shell — you complete it</i>)	The CONFIG console shell. It has the prompt, the settings storage with the R4 defaults, and strict numeric parsing. It also has the window echo and the command dispatch loop over the provided line-reader. You write the fenced stubs. Those stubs are the R4 validators with the range rules and T < E. They also include each message that rejects an entry, and the acquisition hook. The plumbing comes with the starter, and the grade covers the acceptance logic as your work.

Strongly recommended milestone. Make the capture engine see edges and give a plausible histogram. Complete this milestone by the **studio time after the C Proficiency Exam on Sep 9** (Session 5). The histogram gives one bin on the LOOPBACK self-stimulus and a few adjacent bins on the bench generator (§7.4).

Studio 3 (Sep 14) is the last scheduled help before demo day. Use Studio 3 as a bench rehearsal of the capture path, and not as your first work on it. L7 comes on the day that the course assigns P1.

7. Hardware & bench setup

7.1 Signal input

SENSE input	Arduino D5 (connector CN4 pin 6) = PB10 = TIM2_CH3 (AF1)
Pin configuration	Input or AF with a mandatory internal pull-down (GPIOB->PUPDR). A disconnected SENSE lead leaves the line floating . Most analog generators have no output switch, so disconnecting the stimulus lead (as demo step D1 does) is the output switch. Without the pull-down a mid-rail Schmitt input can chatter and defeat POST (R1). The pull-down also holds the line quietly low if the lead comes loose during an acquisition.
Pin tolerance	5 V-tolerant (FT), but apply 0–3.3 V to it (§7.2)
Series protection	1 kΩ resistor in line between the generator lead and D5
Ground	Generator ground clip to a board GND pin

△ **Authoritative pin fact.** D5 = PB10 = TIM2_CH3, AF1. Do not check this fact against third-party copies of the user manual. OCR copies scramble the pin table. The ST document UM2825 Table 19 also contains a known misprint on a different pin (D4). Use this handout and the board schematic.

7.2 Signal generator (device-under-test simulator)

The bench source is an **analog function generator**. The generator has a dial-set frequency, a knob-set amplitude, and a knob-set DC offset, with no presets and no modulation. This imprecision is not a limit on the project, and the project uses it deliberately. The dial of an analog generator is accurate to a few percent only, and the frequency drifts with temperature. An imperfect pacemaker has the same behavior. **The generator is the device under test, and the oscilloscope is the only reference.**

Set each frequency on the demo sheet while you monitor the Period readout of the oscilloscope. Do not trust the dial markings.

Do the setup in this sequence. **Make sure of the signal on the oscilloscope before the SENSE lead touches the board.**

- 1 **Warm-up.** The bench generators operate ≥ 30 minutes before the first demo slot, and they stay on between slots. An analog unit drifts most while it becomes warm. Each TA-published bench number applies to a warm instrument.
- 1 **Square wave.** Set the dial near 1 kHz. Then adjust the dial while you monitor the oscilloscope, until Period reads ≈ 1.000 ms.
- 1 **Levels, with the attenuator in the position that stays.** An analog generator starts in a **bipolar** condition that is symmetric about 0 V. A negative half-cycle violates the -0.3 V absolute minimum of the STM32. Set the output step attenuators to the position that you use for the **full** demo. On many units the generator sums the DC offset before the attenuator, so the offset and the attenuator interact. Then turn the DC-offset knob until the oscilloscope shows **LOW ≈ 0 V and HIGH ≈ 3.0 – 3.3 V**.

Record the attenuator position on your demo sheet. After each change to the amplitude, the attenuator, or the output-range controls, make sure of LOW and HIGH on the oscilloscope again. Do this check **before** you connect the SENSE lead again. A change to a -20 dB attenuator after you set the levels can put tens of volts on the lead. Such a change can also put a negative half-cycle on the lead.

If your generator has a **TTL output** jack, you can use that jack as an alternative. Make sure of the levels on the oscilloscope, as you do for each other source. Some units share that jack with an adjustable **CMOS mode that reaches ~ 15 V**. A HIGH level above **5.5 V** is not permitted, and D0 rejects it. You can apply a true 0–5 V TTL level

to the 5 V-tolerant SENSE pin, through the 1 k Ω series resistor with the board powered. Declare the TTL output at D0.

The amplitude dials are not accurate, and the connected circuit changes the level. Only the oscilloscope indication counts. The level check is a **graded setup item** (D0).

- 1 **Power-on sequence.** USB-power the board **before** you connect the generator lead. The 5 V tolerance of the FT pins applies only when VDD is on.
- 1 **Record your dial and attenuator positions** on the demo sheet. The generator has no presets, and the oscilloscope check at each changeover replaces them.

The demo in §9 uses these stimulus conditions: ~1 kHz square (adjusted on the oscilloscope) · a small dial nudge · a slow dial turn during an acquisition. The other conditions are one fast dial step during an acquisition · ~500 Hz · ~10 kHz · SENSE lead disconnected.

Minimum bench capability. The generator gives a square-wave output from 500 Hz to 10 kHz. The generator has a DC-offset control **or** a TTL output. The oscilloscope does automatic Period measurement with statistics (mean/min/max/ σ). The oscilloscope has a persistence display and a delayed timebase with zoom. **TA pre-work.** The TA publishes a bench addendum before demo week, with values from a warm unit. For each bench generator the TA measures two values. (a) The natural histogram width of a 1000-interval acquisition at ~1 kHz, which is typically a few μ s and gives the D3/D5 expectation. (b) The **total period width** from jitter and drift, minimum to maximum, over ~15 s near 500 Hz, which sizes the D6/D7 values. A generator with a 500 Hz width of more than $\sim\frac{2}{3}$ of the largest legal tolerance (500 μ s) is **not demo-qualified for D7**, and the TA must replace it. Analog units change widely between benches, and the demo sheet cites the TA-published values.

7.3 Do-not-touch pins

The provided drivers exclusively own these pins: **PA1** (OLED SPI clock), **PA7** (OLED MOSI and TLC59731 LED data, shared, jumper JP5 ON by default), **PH0/PC8/PC9** (OLED control), **PH1** (LED select), and **PB6/PB7** (console USART1). A wholesale write to GPIOA→MODER kills your display during the demo. This is the first read-modify-write lesson in the course, and it is not the last one.

7.4 Oscilloscope (the reference)

- Connect CH1 to the generator signal with a BNC tee. Use **DC coupling, a 1 M Ω input, and a checked probe attenuation**. AC coupling hides a bipolar signal behind a display that looks correct. The coupling is part of the graded D0 check.
- Set the edge trigger to ~1.6 V in Normal mode. Use Measure → Period with **statistics** (mean/min/max/ σ). **Clear the statistics after each dial change.** Compare the mean values only when the statistics window of the oscilloscope brackets the acquisition of your instrument. The source drifts, so the two instruments must monitor it at the same time to agree.
- To see the jitter, put the **edge one period after the trigger** on the screen with the delayed timebase or the horizontal reference control. Zoom to 0.5–1 μ s/div, then enable **infinite persistence**. The triggered edge is stable by definition, so persistence at the trigger point shows nothing. The accumulated width on the next edge is the jitter, and you can compare it directly with the width of your histogram.

Metrology honesty. The measurement chain has one reference only, and that reference is the crystal timebase of the oscilloscope. The timebase accuracy is ± 25 –50 ppm, which is ± 0.05 μ s at 1 kHz and twenty times smaller than your 1 μ s bin. The analog generator is **not** a reference. The generator dial is off by a few percent, the cycle-to-cycle jitter spans a few microseconds, and the frequency drifts. These conditions make the generator a good model of a pacemaker under test.

A healthy 1 kHz acquisition gives a **few adjacent bins** and not one bin. Those bins are the true jitter of the generator, and your instrument resolves it. Compare your measured mean with the mean of the oscilloscope, and

not with the dial. Give this statement in your report.

8. Graduate-student requirement (663 sections)

A student in a **graduate section (CMPE 663 / EEEE 663)** SHALL do this extension. The student SHALL demonstrate the extension live at demo row **DG**. The student SHALL give the design and the test results of the extension in the memo. Students in the 563 section are not graded on this section. A team with one 663-section student demonstrates the extension once, and the DG row goes into the demo component of the 663 student only.

GR — DUTY-CYCLE histogram. A true pacing output is a 0.05–2 ms pulse into a ~500 Ω load, and the pulse width is an acceptance parameter with the interval. The duty cycle is the width as a fraction of the period, and it is the shape parameter of the waveform.

- Measure the **duty cycle of each cycle**: $\text{duty} = \text{high_time} / \text{period}$. Use **both-edge capture** on TIM2 channel 3. Configure CCMR and CCER for the two edges. An interrupt SHALL handle the capture, and the ISR SHALL identify the type of each edge (read IDR).
- Report a **duty-cycle histogram with the interval histogram** in the same acquisition report, at rates ≤ 1 kHz. Use **0.1% bins**, and give only the non-empty bins. Each histogram SHALL have its own TOTAL line, and each SHALL account for exactly N.
- Give the minimum, the maximum, and the mean duty cycle to **0.1%**. Use integer arithmetic: $\text{duty_permille} = \text{high_}\mu\text{s} \times 1000 / \text{period_}\mu\text{s}$ (R6 rules apply).

The duty cycle and the interval are independent measurements of the same waveform. A turn of the symmetry control of the generator moves the duty histogram and does not move the interval histogram. The DG demo uses this independence, and the design explanation must state it.

The LOOPBACK stimulus (§6) has a ~50% duty cycle, so it gives a known at-home reference for the extension.

9. Demonstration requirements (the demo sheet)

Each team does the demo **live at the bench** for the instructor or the TA. The demo takes **10 to 20 minutes**, which is the slot length for every project in this course. Two benches operate in alternation, so each team pre-stages on the idle bench during the demo of the team before it. The instructor does the level check and the frequency check with the oscilloscope (D0) again at each changeover. An analog generator has no presets, and the dial positions of the team before you mean nothing.

PuTTY logging stays on during the full demo. The captured PuTTY-log **is** the archived inspection record. The instructor initials the record, and you attach it to your decision memo (§10).

The instructor can do each step again on request. A board reset restarts the full sheet. This rule does not apply to POST, because POST is step D1. Where the sheet gives expected widths or demo tolerances, use the TA-published bench-addendum values for your generator (§7.2).

Pre-stage these items before the slot starts (graded, D0):

- USB-power the board first.
- Put the 1 k Ω series resistor in line.
- Dial the generator to a ~1 kHz square wave with **LOW ≈ 0 V / HIGH ≈ 3.0 –3.3 V checked on the oscilloscope** (or the TTL output declared), with the SENSE lead disconnected.
- Connect the oscilloscope to CH1 with a BNC tee.

- Connect PuTTY and start the logging.
- Make the wiring tidy.

#	Instructor step / stimulus	Required observable result	Verifies
D0	Examine the bench before the start	The pre-stage checklist above is complete. The oscilloscope channel is DC-coupled with a 1 MΩ input, and the probe attenuation is correct . The oscilloscope shows LOW ≥ 0 V and HIGH 3.0–3.6 V <i>before</i> you connect the SENSE lead. The instructor permits about 5 V only with a declared TTL output, and a level above 5.5 V fails. The demo sheet records the attenuator position. A bipolar signal into the pin fails the setup grade	Setup grade
D1	Reset the board with the SENSE lead disconnected . Then connect the lead and push retry	NO DUT SIGNAL on the console and the OLED in ~100 ms, with the retry offer again and again. After you connect the lead and push retry, POST goes into the operator console and shows the CAL-VERIFY defaults (E 1000 / T 50)	R1
D2	Adjust the dial until the mean of the oscilloscope <i>Period</i> reads ≈ 1.000 ms. Clear the statistics after the adjustment and let ~100 counts collect	The student gives the ground-truth chain. The oscilloscope timebase is $\pm 25\text{--}50$ ppm, which is much smaller than the 1 μs bin, so the oscilloscope is the reference. The generator dial is accurate to a few percent only and it drifts, so the generator is the device under test and not a standard	\$7.4

#	Instructor step / stimulus	Required observable result	Verifies
D3	The instructor first tries E = 50 , and the console rejects it (below the E floor) with a range message and a re-prompt. The instructor then tries E = 200 with T = 300 , and the console rejects it (T must be < E). The instructor then accepts the CAL-VERIFY defaults, starts the acquisition while the oscilloscope statistics collect , and reads the oscilloscope mean immediately after the acquisition ends	Blue LED during the ~1.0 s capture, with the static ACQUIRING screen. A histogram of a few adjacent bins around ~1000 μ s, which is the TA-published width for this bench and is the jitter of the generator. TOTAL = 1000 . OUT_LOW = OUT_HIGH = 0. The mean agrees with the Period mean of the oscilloscope in $\pm 2 \mu$s , which is correct because the oscilloscope window brackets the acquisition. The σ agrees with the Period σ of the oscilloscope in 1 μs or 15%, and the larger of the two applies . The rate reads $\approx$ 60,000 ppm, with the R6 calibration-rate statement visible on the console banner. The GREEN PASS latches. The OLED shows the verdict with the minimum and the maximum. The report gives the worst-early and the worst-late deviations and the margin to the $\pm T$ limit, consistent with the printed minimum, maximum, and E (R7). The full cycle from capture to histogram to statistics to verdict completes in 5 s of the start command (R12)	R2 R4 R5 R6 R7 R8 R9 R12
D4	Nudge the dial ~1%, clear the oscilloscope statistics , and do the acquisition again	The histogram cluster and the Period mean of the oscilloscope (accumulated across the acquisition) move together and agree in $\pm 2 \mu$ s. A mis-scaled timebase from the PSC+1 trap reads ~1.6% away <i>from the oscilloscope</i> and cannot give PASS. Return near 1 kHz after this step	R2 R6 R11
D5	Set the delayed timebase and the zoom to 0.5–1 μ s/div. Put the edge one period after the trigger on the screen. Enable infinite persistence. Clear the statistics and let the count get to ~300. Do the acquisition again	The persistence shows a wide next edge, which is the natural jitter of the generator, live. The triggered edge itself stays stable, and the student knows the cause. The histogram width agrees with the TA-published bench value, and all intervals are in-window. TOTAL = 1000. The verdict is PASS . This is the oscillator-noise shape from true hardware and not from a modulation knob	R5 R6 R8

#	Instructor step / stimulus	Required observable result	Verifies
D6	Retune near 500 Hz on the oscilloscope. The student sets E = the period that the oscilloscope reads (≈ 2000) and $T = 50$, which gives a 2 s acquisition. During the acquisition the instructor turns the frequency dial slowly . The instructor permits one more acquisition with no penalty	The histogram becomes wide and walks out of the window. The OUT tallies fill. RED FAIL latches. The OUT_LOW and OUT_HIGH tallies show on the OLED and the console. The TOTAL line sums to 1000, which is the drifting-scheduler shape and proves that the report accounts for every interval while the target moves	R4 R5 R8
D7	The same ~ 500 Hz setup with $T = 500$ and a new acquisition. During the acquisition the instructor makes one fast dial step of ~ 1–3% (with the fine or vernier knob where the unit has one, and the TA addendum gives the target step) and holds it	A bimodal histogram with two separated clusters, and the two clusters are in the $\pm 500 \mu\text{s}$ window. TOTAL = 1000. The student tells the instructor the shape (the mode-switch shape) and why a mean value hides it. The PASS or FAIL verdict depends on where the clusters landed, and the student must defend it from the histogram and not from the LED	R4 R5 R8 R11
D8	Dial to 10.0 kHz, or a small value below it , on the oscilloscope. The period must then read $\geq 100 \mu\text{s}$, which is the E floor of the console. The student sets E = the period that the oscilloscope reads and $T = 10$	The capture takes ~ 0.1 s. TOTAL = exactly 1000 with no missed edge and no double-counted edge at $\sim 100 \mu\text{s}$ between edges. The static ACQUIRING screen stays on the OLED, as R9 gives it	R2 R10
DG (663 sections only)	The student demonstrates the graduate extension at $\leq 1 \text{ kHz}$ with both-edge capture. Where the generator has a symmetry (duty) control , the instructor turns it during a second acquisition	The report gives a duty-cycle histogram with the interval histogram , and each histogram has its own TOTAL of exactly 1000. The duty statistics print to 0.1%. Under the symmetry turn the duty histogram moves and the interval histogram stays, and the student states why the two measurements are independent. The student gives a one-minute explanation of the both-edge design	S8
Q&A	Two questions, for example: <i>why must the instrument not transmit on the UART during a capture (R10)? · why PSC = 63? · why is the oscilloscope the reference and not the generator? · the factory acceptance procedure runs at 30–180 ppm — why does this qualification run at 500 Hz to 10 kHz?</i>	Crisp, correct answers from one team member or the other	Understanding

Time budget. Rehearse this budget.

D0 pre-staged · D1 0:00–1:00 · D2 1:00–2:00 (dial adjustment on the oscilloscope) · D3–D4 2:00–3:30. D5 3:30–4:15 (includes time for the oscilloscope statistics) · D6 4:15–5:00 · D7 5:00–5:45. D7 is the pressure-relief step. The instructor can skip it when the demo runs past its schedule. D8 5:45–6:15. DG 6:15–7:15 (663 sections; a 563 team goes directly to the Q&A). Q&A follows, ~1 minute for each question.

Each acquisition on this sheet completes in 2 s or less, so the sheet fits the slot with time to spare. The full 1000-interval soak at a true pacing rate is deliberately **not** part of this project. The factory procedure documents it.

The demo proves the engine across the full 500 Hz to 10 kHz band, with the drift and step stimuli inside it. A rushed demo shows that the team did not pre-stage the setup.

10. Deliverables

- 1 **Per-person environment smoke test.** This item is a graded checkbox, and it is due before the C Proficiency Exam (S5 Sep 9). Each team member individually builds the unmodified starter, flashes it, and opens PuTTY at their own bench. This work takes 10 minutes, and the provided `loopback.c` with a jumper from D2 to D5 is also the capture-path self-test. A TA checks off this item, and the D0 setup grade includes it (§11).
- 1 **Live demo** as §9 gives it. The instructor or the TA signs the demo sheet.
- 1 **Source code.** Send your project tree to myCourses. The course toolchain must build the tree, and that toolchain is VS Code with the STM32Cube extension. The command `cmake --preset Debug && cmake --build --preset Debug` must succeed from a clean checkout. Do not include work that you did not write. Refer to the provided drivers, and do not send them. Do not include third-party libraries.
- 1 **Decision memo** (PDF, 2–3 pages, team-authored). The memo contains **only** the argued decisions, because the demo proves that the functions operate. The memo has these parts.
 - A title page with the title, the authors, the emails, and the date, and the **areas of focus** of each member.
 - **(a) The display decision** and the cause of the deferred I/O (R9/R10). Argue why the OLED is static during a capture, and what an OLED write costs at the 10 kHz point.
 - **(b) Your σ and the Period-statistics σ of the oscilloscope**, argued. Tell the reader if the two instruments agree, and give the cause.
 - **(c) The probe-compensation cross-check** discussion. Measure the ~1 kHz probe-compensation output of the oscilloscope, and compare its histogram with the histogram of the generator. Tell the reader which source makes the better pacemaker model, and give the cause. No source here is a timing standard, because the oscilloscope timebase is the only reference in the two measurements.
 - **(d) Short lessons learned**, and honest ones. Students learned "the prescaler is PSC+1" here before.
 - 663 students: one page on the graduate extension, with the both-edge design and the duty-cycle test results (§8).

Attachments, not prose. Attach the instructor-initialed demo sheet and the PuTTY-log. These two items ARE the official results record. The memo has no test-plan section, no results-vs-expected section, and no mandatory block diagram.

11. Grading

Component	Weight	Notes
Demo — tidy, correct bench setup (D0)	10%	Levels checked on the oscilloscope (no bipolar signal into the pin), supply sequence, series R, logging. Includes the pre-S5 smoke-test checkbox of each member (refer to §10 item 0)
Demo — demo-sheet functions complete (D1–D8)	40%	The instructor initials each row
Demo — operates without faults or restarts	10%	This row does not include the POST demo. A hang or a reflash forfeits this row. One carve-out applies. If a lead accident interrupts an acquisition, the instructor permits a board reset and a re-run of that step, with no forfeit (R12 scope note)
Program design	15%	Correct initialization, function decomposition, no copy-paste-edit blocks, hardware and software separation
Source code structure & readability	10%	
Decision memo	15%	As §10 gives it
Graduate requirement (DG) — 663 sections only	inside the 40% demo-rows component	For a 663 student, row DG joins rows D1–D8 in the 40% demo-rows component, and DG gets a grade like each other demo row. A missing DG or a defective DG forfeits its share of that component. The maximum score is the same for the two sections. Not applicable to 563 students.

The graduate (663) differentiation on this project is the §8 requirement, at demo row DG. The two sections share all other work. More graduate extensions continue on Projects 2–5, as the syllabus gives them.

Legacy solutions. The ancestor of this assignment was at RIT for two decades, and complete legacy solutions are public on GitHub. Know the limits of those solutions. They **hang forever on out-of-window signals**, and demo step D6 finds that behavior. They have no OUT tallies and no deferred-I/O discipline. They have a well-known stale-first-timestamp defect, which R2 now forbids. Demo step D3 finds it, because the garbage first interval lands in OUT_HIGH and D3 requires OUT_HIGH = 0 with a PASS verdict. The course designed the demo sheet so that inherited code fails visibly.

Cite each source that you refer to. Send only the work that you wrote.

12. References

- STM32WB55 Reference Manual **RM0434** (TIM2 §chapters: CR1, PSC, EGR, CCMR2, CCER, SR, CCR3)
- STM32WB55 Datasheet **DS11929** (pin AF tables, FT-pin tolerances)
- **UM2825** STM32WB5MM-DK user manual, for the board schematic references. Table 19 has a D4 misprint, and the pin facts in this handout are authoritative for the course.
- Board BSP source (pin ownership of provided drivers): github.com/STMicroelectronics/STM32CubeWB_Drivers/BSP/STM32WB5MM-DK
- *Timer Setup for Capture* — course handout, ships with this assignment
- The manual of your bench generator. Find the DC-offset controls, the attenuator controls, and the symmetry (duty) control before D0. Do not apply a bipolar signal to the board.

- Background material (optional, for the scenario context). True pacemaker QA works much as this specification gives it. Refer to ISO 14708-2, the implantable pacemaker standard. The company is fictional.